AQ6.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

- [ ] Nullity and rank of the identity transformation on a vector space of dimension n are 0 and n respectively.
- [ ] Nullity and rank of the identity transformation on a vector space of dimension n are 1 and $n-1$ respectively.
- [ ] Nullity and rank of the identity transformation on a vector space of dimension n are n and 0 respectively.
- [ ] Nullity and rank of an isomorphism between two vector spaces V and W (both of dimension n) are n and 0 respectively.
- [ ] Nullity and rank of an isomorphism between two vector spaces V and W (both of dimension n) are 0 and n respectively.
- [ ] There cannot exist an isomorphism between two vector spaces whose dimensions are not the same.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Nullity and rank of the identity transformation on a vector space of dimension n are 0 and n respectively.

Nullity and rank of an isomorphism between two vector spaces V and W (both of dimension n) are 0 and n respectively.

There cannot exist an isomorphism between two vector spaces whose dimensions are not the same.

Consider the following linear transformation:

$T: R^3 \to R^3$

$T(x, y, z) = (x - z, 2x + 3y + z, 3y + 3z)$

Answer questions 2,3,4, and 5, using the information given above.

1 point

Which of the matrices corresponds to the given linear transformation T with respect to the standard ordered basis of R^3 for both the domain and the co-domain?

- ( ) $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$
- ( ) $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$

○

$$\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$$

○

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

1 point

What will be the kernel of T?

○ $Span(S)$, where $S = \{(x, 0, z), (0, y, z) \mid x, y, z \in R\}$.

○ $Span(S)$, where $S = \{(z, 0, z), (0, -z, z) \mid z \in R\}$.

○ $Span(S)$, where $S = \{(z, -z, z) \mid z \in R\}$.

○ $Span(S)$, where $S = \{(x, 0, z), (0, -x, z) \mid x, z \in R\}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$Span(S)$, where $S = \{(z, -z, z) \mid z \in R\}$.

What will be the $rank(T)$?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2

1 point

1 point

Choose the correct option.

○ T is an isomorphism.

○ T is one to one but not onto.

○ T is onto but not one to one.

○ T is neither one to one, nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is neither one to one, nor onto.

Let $T: R^4 \to R^4$ be a linear transformation. Suppose $v_1 \neq v_2$ are two distinct solutions of $T(x) = 0$. What is the maximum value that rank(A) can take?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Level 2

1 point
Choose the set of correct options.

- [ ] Any injective linear transformation between any two vector spaces which have the same dimensions, must be an isomorphism.
- [ ] Any surjective linear transformation between any two vector spaces which have the same dimensions, must be an isomorphism.
- [ ] There does not exist any surjective linear transformation from R^2 to R^3.
- [ ] There does not exist any injective linear transformation from R^2 to R^3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Any injective linear transformation between any two vector spaces which have the same dimensions, must be an isomorphism.
Any surjective linear transformation between any two vector spaces which have the same dimensions, must be an isomorphism.
There does not exist any surjective linear transformation from R^2 to R^3.
1 point
Let T be an injective linear transformation as follows :

$T: R^3 \to R^3$ such that, $T(1, 0, 0) = (0, 1, 1)$ and $T(0, 1, 0) = (1, 0, 1)$

Which of the following can be a possible definition of T?

- ( ) $T(x, y, z) = (y, x, x + y)$
- ( ) $T(x, y, z) = (y + z, x + 2z, x + y + 3z)$
- ( ) $T(x, y, z) = (y + z, x + 2z, x + y + 4z)$
- ( ) $T(x, y, z) = (y, x + y, x + y + z)$

No, the answer is incorrect.
Score: 0

Accepted Answers:
$T(x, y, z) = (y + z, x + 2z, x + y + 4z)$

1 point
Consider the linear transformation

$M_{2\times 2}(\mathbb{R}) \to M_{2\times 2}(\mathbb{R})$ such that $T(A) = PA$, where $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and a, b, c, d are in $\mathbb{R}$.

Let $\beta = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ be an ordered basis of $M_{2\times 2}(\mathbb{R})$. Which of the following matrices represents the matrix corresponding to the linear transformation T with respect to β for both domain and co-domain?

- ○ $\begin{bmatrix} a & b & 0 & 0 \\ 0 & 0 & c & d \\ a & b & 0 & 0 \\ 0 & 0 & c & d \end{bmatrix}$

- ○ $\begin{bmatrix} a & 0 & b & 0 \\ c & 0 & d & 0 \\ c & a & 0 & b \\ 0 & c & 0 & d \end{bmatrix}$

- ○ $\begin{bmatrix} a & 0 & 0 & a \\ c & 0 & 0 & c \\ b & 0 & 0 & b \\ d & 0 & 0 & d \end{bmatrix}$

- ○ $\begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix}$

1 point
Choose the set of correct options.

- [ ] There exists a linear transformation $T: R^2 \rightarrow R^2$ such that Image(T)= Kernel(T).
- [ ] There exists a linear transformation $T: R^3 \rightarrow R^3$ such that Image(T)= Kernel(T).
- [ ] If $T: V \rightarrow V$ is a linear transformation and $v_1, v_2 \in V$ are linearly independent then $T(v_1)$, $T(v_2)$ are also linearly independent.
- [ ] If $T: V \rightarrow V$ is a linear transformation and $T(v_1)$, $T(v_2)$ are linearly independent then $v_1, v_2 \in V$ are also linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

There exists a linear transformation $T: R^2 \rightarrow R^2$ such that Image(T)= Kernel(T).
If $T: V \rightarrow V$ is a linear transformation and $T(v_1)$, $T(v_2)$ are linearly independent then $v_1, v_2 \in V$ are also linearly independent.

Check Answers
Your score is: 0/10